



$\Leftrightarrow$

$$\begin{aligned}
 C^{[0]} \bar{C}^{[1]} \hat{U}^{[2]} &= |0\rangle\langle 0|^{[0]} \otimes \hat{I}^{[1,2]} + |1\rangle\langle 1|^{[0]} \otimes \bar{C}^{[1]} \hat{U}^{[2]} \\
 &= \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes |0\rangle\langle 0| \otimes (\hat{U} - \hat{I})
 \end{aligned}$$